



Unleash your potential

Master Presenter's Manual

for

Full Stack MERN Development

Curriculum Code: OV-7110

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1. Introduction

This Presenter's Manual is to be used for Full Stack MERN Development curriculum. The course aims at introducing students to full-fledged Website development. This course also introduces students to perform programming using ReactJS, MongoDB, and design and develop full stack applications using NodeJS and Express.js.

The student will learn the following modules in this course:

- Building Modern Websites
- UI/UX for Responsive Web Design
- Frontend Web Development with React
- Material UI
- Distributed Version Control
- Large Data Management (MongoDB)
- Server-side Development
- Building Full Stack Applications
- Project Guide-Full Stack MERN Application

The course contents are specially designed for those who are keen on getting themselves a career in the field of full stack application development.

After the completion of this course, students will be able to:

- Use HTML 5 /CSS 3 /JavaScript to implement well structured Websites
- Learn the basic principles of effective Web UX/UI design
- Take your project to the next level with Material UI
- Become an expert front end developer using ReactJs
- Learn the basics of Git for tracking, branching, merging, and managing code revisions
- Write server-side applications using JavaScript framework Node. Js
- Start learning how to build real world full stack apps
- Develop a full stack Web application using MERN

The Project in this term will involve developing a full stack application using MongoDB NodeJS, Express.js, and ReactJS.

2. General Information on Session Schedule

The table below shows the details of the modules, the number of hours being spent on each module along with a break-up of the hours being spent on each module.

This course is based on theory and practice lab session.

A session is of **TWO** hours duration.

Module Wise Session Allocation

Module	Theory Hrs.	Online Hrs.	Total Hrs.	Self-study Hours	Skill Code
Building Modern Websites	20	20	40	12	177
UI/UX for Responsive Web Design	8	8	16	6	368
Frontend Web Development with React	12	12	24	8	1337
Material UI	6	0	6	0	1316
Distributed Version Control	6	6	12	8	1273
Large Data Management (MongoDB)	16	16	32	10	399
Server-side Development	12	12	24	12	396
Building Full Stack Applications	20	20	40	8	1317
Project Guide-Full Stack MERN Application	2	0	2	0	1317
Total Hours	102	94	196	64	

The table below shows the details of the modules, the number of hours being spent on each module along with a break-up of the hours being spent on each session. Refer to the respective module presenter's manuals to understand further details of each session.

Topic\Week		1	2	3	4	5	6	7	8	9	10	11	12	13
Building Modern Websites	D1	TL1	TL5	TL9	TL13	TL17								
	D2	TL2	TL6	TL10	TL14	TL18								
	D3	TL3	TL7	TL11	TL15	TL19								
	D4	TL4	TL8	TL12	TL16	TL20								
		S1-S2		S3-S4		S5-S6								
UI/UX for Responsive Web Design	D1						TL1	TL5	IT1					
								S2						
	D2						TL2	TL6						
							S1							
	D3						TL3	TL7						
	D4						TL4	TL8						
Frontend Web Development with React	D1									TL4	TL8	TL12		
										S1-S2		S3-S4		
	D2								TL1	TL5	TL9			
	D3								TL2	TL6	TL10			
	D4								TL3	TL7	TL11			
Material UI	D1													
	D2											T1		
	D3											T2		
	D4											T3		
Distributed Version Control	D1												TL1	TL5
	D2												TL2	TL6
														S3-S4
	D3												TL3	IT2
	D4												TL4	
													S1-S2	

Topic\Week		13	14	15	16	17	18	19	20	21	22	23	26	27	28
Large Data Management (MongoDB)	D1		TL2	TL6	TL10	TL14									
	D2		TL3	TL7	TL11	TL15									
				S1-S3											
	D3		TL4	TL8	TL12	TL16									
						S4-S5									
	D4	TL1	TL5	TL9	TL13										
Server-side Development	D1						TL2	TL6	TL10						
								S1-S3							
	D2						TL3	TL7	TL11						
	D3						TL4	TL8	TL12						
									S4-S6						
	D4					TL1	TL5	TL9							
Building Full Stack Applications	D1									TL2	TL6	TL10	TL14	TL18	
	D2									TL3	TL7	TL11	TL15	TL18	
	D3									TL4	TL8	TL12	TL16	TL20	
														S3-S4	
	D4								TL1	TL5	TL9	TL13	TL17	IT3	
												S1-S2			
Project Guide-Full Stack MERN Application	D1														
	D2														PRJ
	D3														
	D4														

Where,
D = Day
T = Theory Session, TL = Online Session
IT = Internal Test
PRJ = Project
S = Self-study Hours (over and above the course duration)

Note:

The Exam duration is over and above the course duration.

3. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ Deliverables for Students:

Module	Deliverable
Kit Details	Kit Code: OV-7110-KIT01-01-I Kit Name: Kit of 9 titles for OV-7110 Full Stack MERN
Building Modern Websites	Book Code: AINNFENDT10323E000 Book Name: FrontEnd Technologies for Beginners
UI/UX for Responsive Web Design	Book Code: AINNEUIUX10721E000 Book Name: Elegant and Effective Web Design with UI and UX
Frontend Web Development with React	Book Code: AINNREACT10423E000 Book Name: Frontend Web Development with ReactJS
Material UI	Book Code: AINNMATUI10323E000 Book Name: Material UI
Distributed Version Control	Book Code: AINNGITHB11221E000 Book Name: Version Control with Git and GitHub
Large Data Management (MongoDB)	Book Code: AINNMGDB10823E000 Book Name: Managing Large Databases Using MongoDB
Server-side Development	Book Code: AINNDEJS10124E000 Book Name: Server-side Development with NodeJS
Building Full Stack Applications	Book Code: AINNBFSK10523E000 Book Name: Building Full Stack Applications
Project Guide-Full Stack MERN Application	Book Code: AINNPGMRN10124E000 Book Name: PG-Full Stack MERN Application

➤ Deliverables for Faculty

Module	Deliverable
Kit Details	Kit Code: OV-7110-KIT01-01-I Kit Name: Kit of 9 titles for OV-7110 Full Stack MERN
Building Modern Websites	Book Code: AINNFENDT10323E000 Book Name: FrontEnd Technologies for Beginners
UI/UX for Responsive Web Design	Book Code: AINNEUIUX10721E000 Book Name: Elegant and Effective Web Design with UI and UX
Frontend Web Development with React	Book Code: AINNREACT10423E000 Book Name: Frontend Web Development with ReactJS
Material UI	Book Code: AINNMATUI10323E000 Book Name: Material UI
Distributed Version Control	Book Code: AINNGITHB11221E000 Book Name: Version Control with Git and GitHub
Large Data Management (MongoDB)	Book Code: AINNMGDB10823E000 Book Name: Managing Large Databases Using

Module	Deliverable
	MongoDB
Server-side Development	Book Code: AINNDEJS10124E000 Book Name: Server-side Development with NodeJS
Building Full Stack Applications	Book Code: AINNBFSK10523E000 Book Name: Building Full Stack Applications
Project Guide-Full Stack MERN Application	Book Code: AINNPGMRN10124E000 Book Name: PG-Full Stack MERN Application
	Master PM Module PMs Classroom Presentations (wherever applicable) Trainer Guide (wherever applicable)

OnlineVarsity Deliverables

OnlineVarsity is a digital content access and collaboration platform for Aptech students. The numerous interesting features of the platform will enhance learning experience of the students.

(Faculties can access eBooks and other components in OnlineVarsity through Aprack.)

Resources available on OnlineVarsity (as applicable):

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Show Me How - Student can view a step-wise simulation/demonstration of the module related topics.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn - Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Internal Examination
- Final Examination
- Project

4.1 Award of Grades

Overall Weighted Marks $\geq 40\%$ but $< 60\%$ qualifies for **PASS**

Overall Weighted Marks $\geq 60\%$ but $< 75\%$ qualifies for **CREDIT and**

Overall Weighted Marks $\geq 75\%$ qualifies for **DISTINCTION**

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in each of the elements of Internal Examination as well as Final Examination, otherwise he/she will be declared as 'Referred'.

4.2 Internal Examination

Below is the list of modules to be covered in Internal Tests along with their weightages:

Internal Test 1 (IT1)

Marks: 20

Duration: 25 minutes

Type: Objective

S#	Name of the Subject	Marks
1.	Building Modern Websites	14
2.	UI/UX for Responsive Web Design	6

Internal Test 2 (IT2)

Marks: 20

Duration: 25 minutes

Type: Objective

S#	Name of the Subject	Marks
1.	Frontend Web Development with React	10
2.	Material UI	3
3.	Distributed Version Control	7

Internal Test 3 (IT3)

Marks: 20

Duration: 25 minutes

Type: Objective

S#	Name of the Subject	Marks
1.	Large Data Management (MongoDB)	6
2.	Server-side Development	4
3.	Building Full Stack Applications	10

4.3 Final Examination 1 (E1)

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination 1 (E1)

Marks: 50

Duration: 60 minutes

Type: Objective

S#	Name of the Subject	Marks
1.	Large Data Management (MongoDB)	15
2.	Server-side Development	10
3.	Building Full Stack Applications	15
4.	Building Modern Websites	10

Final Examination 2 (R1)

Marks: 50

Duration: 120 minutes

Type: Practical

S#	Name of the Subject	Marks
1.	Large Data Management (MongoDB)	15
2.	Server-side Development	10
3.	Building Full Stack Applications	25

Final Project (P)

Marks: 100

Duration: 60 minutes

Type: Project

S#	Name of the Subject	Percentage of Content
1.	Project Guide-Full Stack MERN Application	100%

Evaluation Details

OES-70%

Pract-10%

Test-10%

eProject-10%

5. Classroom and Lab Pre-requisites

The requirements for conducting the sessions of the new course have been described as follows:

5.1 Software Requirements

Microsoft Windows 10 or higher
CoffeeCup Free HTML5 Editor
Figma Toolkit
React 18.x or higher
GitHub Desktop 3.x
MongoDB 6.X
Node.js 19.x
Express.js 5.x or higher, Node.js 19.x or higher, MongoDB 6.x

5.2 Recommended Hardware

Intel Core i5 Processor or higher
8 GB RAM or above
Color SVGA
500 GB Hard Disk space
Mouse
Keyboard

5.3 Other Requirements (For Classroom)

Computer for giving demonstrations wherever applicable
LCD projector

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